FFT

Evaluate $A(x) = 1+ 2x+3x^2+4x^3$ using FFT

1) Setup — choose n and roots of unity

Here $n = 4$ (we have 4 coefficients).

The 4th roots of unity are:

$$\omega_4^0 = 1, \quad \omega_4^1 = i, \quad \omega_4^2 = -1, \quad \omega_4^3 = -i.$$

Goal: compute $A(\omega_4^k)$ for $k = 0, 1, 2, 3$.

2) Split into even and odd polynomials

Take even-indexed coefficients (indices 0,2) and odd-indexed (1,3):

- $A^{[0]}(x)$ (even indices): $a_0 + a_2 x = 1 + 3x$.
- $A^{[1]}(x)$ (odd indices): $a_1 + a_3 x = 2 + 4x$.

So

$$A(x) = A^{[0]}(x^2) + x \cdot A^{[1]}(x^2).$$

(That's the divide step.)

3) Recursively evaluate the two size-2 polynomials

By the **halving lemma**, to evaluate A at ω_4^k we need $A^{[0]}$ and $A^{[1]}$ at $(\omega_4^k)^2 = \omega_2^k$.

The 2nd roots of unity are 1 and -1. So we evaluate each small polynomial at $x = 1$ and $x = -1$.

Compute values precisely:

For $A^{[0]}(x) = 1 + 3x$

- $A^{[0]}(1) = 1 + 3 \cdot 1 = 4.$
- $A^{[0]}(-1) = 1 + 3 \cdot (-1) = 1 - 3 = -2.$

For $A^{[1]}(x) = 2 + 4x$

- $A^{[1]}(1) = 2 + 4 \cdot 1 = 6.$
- $A^{[1]}(-1) = 2 + 4 \cdot (-1) = 2 - 4 = -2.$

So the "half-size" FFT results are:

- $\text{Even-values } Y_{\text{even}} = [4, \; -2]$ (for k=0,1)
- $\text{Odd-values } Y_{\text{odd}} = [6, \; -2]$ (for k=0,1)

(That finished the recursion — base case is evaluating degree-0 polynomials or directly computing these small values.)

4) Combine the half results to get full FFT (the merge)

Use the FFT combine formulas for $n = 4$:

For $k = 0, 1$:

$$Y[k] = Y_{\text{even}}[k] + \omega_4^k \cdot Y_{\text{odd}}[k]$$

$$Y[k+2] = Y_{\text{even}}[k] - \omega_4^k \cdot Y_{\text{odd}}[k]$$

Now compute both k values.

k = 0 (here $\omega_4^0 = 1$; the ω_2^0 evaluations we used were at x=1)

- $Y[0] = 4 + 1 \cdot 6 = 10.$
- $Y[2] = 4 - 1 \cdot 6 = -2.$

k = 1 (here $\omega_4^1 = i$; the ω_2^1 evaluations we used were at x=-1)

- $Y[1] = (-2) + i \cdot (-2) = -2 - 2i.$
- $Y[3] = (-2) - i \cdot (-2) = -2 + 2i.$

(Notice the second-half values come from the same $Y_{\text{even}}[k]$ and $Y_{\text{odd}}[k]$ with a minus — that's the **cancellation symmetry**.)